

CHARITY AND AID MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

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Course	AIT 725 – Spring 2026
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Version	v1.3 – Final
Date	March 2026

Revision History

Version	Date	Author	Description of Changes
1.0	01/04/2026	Shivani Pothkanuri	Initial draft submitted
1.1	02/15/2026	Shivani Pothkanuri	Draft v1.1 – advisor review incorporated; FRs expanded
1.2	03/01/2026	Shivani Pothkanuri	Draft v1.2 – user stories, workflow diagrams, ERD added
1.3	03/21/2026	Shivani Pothkanuri	Final version – all advisor feedback applied; stack updated to Java/Spring Boot/MySQL

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1. System Overview / Background

1.1 Purpose

The purpose of the Charity and Aid Management System (CAMS) is to design and develop a web-based platform that centralizes donation management, campaign management, beneficiary registration, and aid distribution processes for small and mid-sized charitable organizations. Currently, many charitable organizations rely on spreadsheets, emails, and manual paper records to manage daily operations, resulting in data duplication, inconsistencies, delays, and a lack of transparency for stakeholders. This system aims to replace fragmented manual processes with a single, unified platform that improves operational efficiency, enhances donor trust, and ensures structured and accountable aid distribution. The platform will serve multiple user roles including donors, organizational staff, case managers, system administrators, and organizational leadership, each with defined access and functional responsibilities.

1.2 Scope

The Charity and Aid Management System will be developed as a three-tier web application using HTML5 and Bootstrap 5 for the presentation layer, Java 17 with Spring Boot 3 for the application logic layer, and MySQL 8 for the data storage layer. The system will support donor registration, campaign creation and management, monetary and in-kind donation recording, beneficiary registration, aid request submission and approval workflows, aid fulfillment tracking, inventory management, role-based access control, and summary reporting with dashboard views. Security will be enforced through Spring Security with JWT-based authentication and bcrypt password hashing. The system will be deployed as a semester prototype hosted on AWS EC2 or Render.com, accessible through modern web browsers. Out of scope are live payment gateway integrations, mobile native applications, and integration with external charity networks.

1.3 System Context

The Charity and Aid Management System operates as a centralized web-based platform connecting five primary user types: Donors, who browse campaigns and submit donations; Organizational Staff, who manage campaigns, register beneficiaries, and record aid requests; Case Managers, who review and approve or deny aid requests; System Administrators, who manage user accounts, roles, permissions, and system configuration; and Organizational Leadership, who monitor performance through dashboards and reports. The system interacts externally with an email notification service for automated communications and a MySQL 8 database for persistent data storage. All client-server communication is secured via HTTPS and authenticated using JWT tokens managed by Spring Security with role-based access control enforced through `@PreAuthorize` annotations.

1.4 Key Features

The Charity and Aid Management System provides the following key capabilities:

- **Campaign Management:** Staff can create, update, close, and monitor fundraising campaigns including goals, timelines, and categories.
- **Donation Tracking:** Donors can record monetary and in-kind donations, download receipts, and view their full donation history.
- **Beneficiary Management:** Staff can register beneficiaries and maintain profiles and complete aid history.
- **Aid Request Workflow:** Staff log aid requests; case managers review, approve, or deny them with documented justification.
- **Aid Fulfillment Tracking:** Approved requests are tracked through delivery with inventory deduction and fulfillment timestamps.
- **Inventory Management:** In-kind donation items are tracked with availability checks enforced before fulfillment.
- **Role-Based Access Control:** Spring Security enforces role-specific permissions for all five user types.
- **Reporting and Dashboards:** Administrators and leadership can generate operational and management reports with KPI dashboards.
- **Audit Trail:** All record modifications, approvals, denials, and login events are logged for full accountability.

2. Requirements

2.1 Functional Requirements

This section defines the complete set of functional requirements for the Charity and Aid Management System. Each requirement is expressed as a single, testable statement describing one specific capability the system must provide. Requirements are grouped by functional area and assigned a unique identifier for traceability. Priority levels indicate relative importance to core system operation. Every data entity is covered with full CRUD (Create, Read, Update, Delete) support as applicable. A total of 94 functional requirements are defined across ten functional areas, ensuring comprehensive coverage of all system behavior.

2.1.1 User Management

ID	Feature Area	Requirement Statement	Priority
FR-01	User Management	The system shall allow a new donor to self-register by providing name, email, phone, and password.	High
FR-02	User Management	The system shall allow an administrator to register a new staff, case manager, or leadership user.	High
FR-03	User Management	The system shall authenticate all users using a registered email and password credential pair.	High
FR-04	User Management	The system shall allow administrators to assign an initial role to a newly registered user.	High
FR-05	User Management	The system shall allow administrators to update the role of an existing user.	High
FR-06	User Management	The system shall allow administrators to deactivate a user account.	Medium
FR-07	User Management	The system shall allow administrators to reactivate a previously deactivated user account.	Medium
FR-08	User Management	The system shall allow administrators to temporarily suspend a user account.	Medium
FR-09	User Management	The system shall automatically lock a user account after 5 consecutive failed login attempts.	High
FR-10	User Management	The system shall allow a user to reset their password via a secure email verification link.	High
FR-11	User Management	The system shall enforce multi-factor authentication for all users assigned the Administrator role.	High
FR-12	User Management	The system shall log each user login event with the user ID and timestamp.	High
FR-13	User Management	The system shall log each user logout event with the user ID and timestamp.	High
FR-14	User Management	The system shall allow a user to view their own profile information.	High

FR-15	User Management	The system shall allow a user to update their own name and phone number.	High
FR-16	User Management	The system shall restrict deletion of user accounts; deactivation is the only permitted removal action.	High

2.1.2 Donor Management

ID	Feature Area	Requirement Statement	Priority
FR-17	Donor Management	The system shall allow donors to browse a paginated list of all active campaigns.	High
FR-18	Donor Management	The system shall allow donors to view the full details of a selected campaign.	High
FR-19	Donor Management	The system shall allow donors to submit a monetary donation linked to an active campaign.	High
FR-20	Donor Management	The system shall allow donors to submit an in-kind donation linked to an active campaign.	High
FR-21	Donor Management	The system shall generate a unique transaction ID for each recorded donation.	High
FR-22	Donor Management	The system shall generate a downloadable PDF receipt for each completed donation.	High
FR-23	Donor Management	The system shall allow donors to view their complete donation history.	High
FR-24	Donor Management	The system shall allow donors to filter their donation history by date range.	High
FR-25	Donor Management	The system shall allow donors to filter their donation history by campaign.	High
FR-26	Donor Management	The system shall allow donors to schedule a recurring donation to an active campaign.	Medium
FR-27	Donor Management	The system shall allow donors to cancel a scheduled recurring donation.	Medium
FR-28	Donor Management	The system shall prevent a donor from submitting a donation to a closed or completed campaign.	High
FR-29	Donor Management	The system shall send an automated email confirmation to the donor after recording a donation.	High

2.1.3 Campaign Management

ID	Feature Area	Requirement Statement	Priority
FR-30	Campaign Management	The system shall allow authorized staff to create a new fundraising campaign.	High
FR-31	Campaign Management	The system shall allow authorized staff to update the title, description, or dates of an existing campaign.	High

FR-32	Campaign Management	The system shall allow authorized staff to update the goal amount of an existing campaign.	High
FR-33	Campaign Management	The system shall allow authorized staff to close an active campaign manually.	Medium
FR-34	Campaign Management	The system shall allow authorized staff to delete a campaign that has no associated donations.	Medium
FR-35	Campaign Management	The system shall allow authorized staff to assign a category to a campaign.	Medium
FR-36	Campaign Management	The system shall track the running total of donations collected against the campaign goal.	High
FR-37	Campaign Management	The system shall display a percentage progress indicator for each campaign.	High
FR-38	Campaign Management	The system shall automatically mark a campaign status as Completed when the goal amount is reached.	High
FR-39	Campaign Management	The system shall notify authorized staff via email when a campaign reaches 90% of its goal.	Medium
FR-40	Campaign Management	The system shall allow staff to view a list of all campaigns with their current status.	High

2.1.4 Beneficiary Management

ID	Feature Area	Requirement Statement	Priority
FR-41	Beneficiary Management	The system shall allow authorized staff to register a new beneficiary with name, phone, and address.	High
FR-42	Beneficiary Management	The system shall allow authorized staff to update a beneficiary's contact information.	High
FR-43	Beneficiary Management	The system shall allow authorized staff to view a beneficiary's profile.	High
FR-44	Beneficiary Management	The system shall allow authorized staff to view a beneficiary's complete aid request history.	High
FR-45	Beneficiary Management	The system shall allow authorized staff to deactivate a beneficiary record.	Medium
FR-46	Beneficiary Management	The system shall prevent creation of a duplicate beneficiary record based on matching name and phone number.	High

2.1.5 Aid Request Management

ID	Feature Area	Requirement Statement	Priority
FR-47	Aid Request Management	The system shall allow authorized staff to record a new aid request for a registered beneficiary.	High
FR-48	Aid Request Management	The system shall allow authorized staff to specify the aid type for a new request.	High

FR-49	Aid Request Management	The system shall allow authorized staff to update an aid request that is still in Pending status.	High
FR-50	Aid Request Management	The system shall allow case managers to view a list of all pending aid requests.	High
FR-51	Aid Request Management	The system shall allow case managers to view the full details of a specific aid request.	High
FR-52	Aid Request Management	The system shall allow case managers to approve a pending aid request.	High
FR-53	Aid Request Management	The system shall allow case managers to deny a pending aid request.	High
FR-54	Aid Request Management	The system shall require case managers to enter a justification note when denying an aid request.	High
FR-55	Aid Request Management	The system shall allow case managers to escalate a pending request to an administrator.	Medium
FR-56	Aid Request Management	The system shall record the decision timestamp for every approval or denial action.	High
FR-57	Aid Request Management	The system shall track aid request status through four states: Pending, Approved, Denied, Fulfilled.	High
FR-58	Aid Request Management	The system shall prevent duplicate aid requests for the same beneficiary and aid type within a 30-day window.	High
FR-59	Aid Request Management	The system shall require mandatory supporting documentation before approving a high-value aid request.	High
FR-60	Aid Request Management	The system shall notify the beneficiary by email when their aid request is approved.	High
FR-61	Aid Request Management	The system shall notify the beneficiary by email when their aid request is denied.	High
FR-62	Aid Request Management	The system shall display the full status history of an aid request including timestamps and actor.	High
FR-63	Aid Request Management	The system shall record which staff member submitted each aid request.	High
FR-64	Aid Request Management	The system shall record which case manager approved or denied each aid request.	High

2.1.6 Aid Fulfillment

ID	Feature Area	Requirement Statement	Priority
FR-65	Aid Fulfillment	The system shall allow authorized staff to record the fulfillment of an approved aid request.	High
FR-66	Aid Fulfillment	The system shall record the fulfillment date and the staff member who completed the fulfillment.	High

FR-67	Aid Fulfillment	The system shall prevent fulfillment of an aid request that has not been approved.	High
FR-68	Aid Fulfillment	The system shall prevent fulfillment if inventory is insufficient for in-kind aid requests.	High
FR-69	Aid Fulfillment	The system shall deduct the fulfilled item quantities from inventory upon recording fulfillment.	High
FR-70	Aid Fulfillment	The system shall send an automated notification to the assigned case manager upon fulfillment.	Medium

2.1.7 Inventory Management

ID	Feature Area	Requirement Statement	Priority
FR-71	Inventory Management	The system shall allow authorized staff to create a new inventory item record.	Medium
FR-72	Inventory Management	The system shall allow authorized staff to update the quantity of an existing inventory item.	High
FR-73	Inventory Management	The system shall allow authorized staff to view the complete inventory list with quantities.	High
FR-74	Inventory Management	The system shall allow authorized staff to delete an inventory item that has zero quantity and no pending aid requests.	Medium
FR-75	Inventory Management	The system shall allow tracking of an expiry date for perishable inventory items.	Medium
FR-76	Inventory Management	The system shall trigger a system alert when an inventory item quantity falls below its defined low threshold.	Medium

2.1.8 Reporting and Dashboards

ID	Feature Area	Requirement Statement	Priority
FR-77	Reporting	The system shall generate a donation summary report showing total donations by campaign and period.	High
FR-78	Reporting	The system shall generate an active campaigns report showing each campaign's progress toward its goal.	High
FR-79	Reporting	The system shall generate a pending aid requests report listing all unresolved requests.	High
FR-80	Reporting	The system shall generate a fulfilled aid report showing completed requests and resources used.	High
FR-81	Reporting	The system shall generate a compliance report based on the system audit log.	Medium
FR-82	Reporting	The system shall provide a KPI dashboard for leadership showing total donations, campaign progress, open vs fulfilled requests, and resource availability.	High

FR-83	Reporting	The system shall allow administrators to filter all reports by date range.	High
FR-84	Reporting	The system shall allow administrators to filter donation reports by donor.	High
FR-85	Reporting	The system shall allow administrators to filter donation reports by campaign.	High
FR-86	Reporting	The system shall allow exporting of donation reports in CSV format.	Medium
FR-87	Reporting	The system shall restrict all report and dashboard access to authorized roles only.	High

2.1.9 Audit and Transparency

ID	Feature Area	Requirement Statement	Priority
FR-88	Audit & Transparency	The system shall maintain an audit trail recording all create, update, and delete operations on key records.	High
FR-89	Audit & Transparency	The system shall restrict the deletion of donation and financial transaction records.	High
FR-90	Audit & Transparency	The system shall allow administrators to view the complete audit log.	High

2.1.10 Notifications and Communication

ID	Feature Area	Requirement Statement	Priority
FR-91	Notifications	The system shall send automated reminder notifications to case managers for requests pending approval for more than 48 hours.	Medium
FR-92	Notifications	The system shall allow administrators to configure notification email templates.	Medium
FR-93	Notifications	The system shall allow staff to send bulk campaign update emails to all donors linked to a campaign.	Medium
FR-94	Notifications	The system shall notify organizational leadership when a single donation exceeds a configured high-value threshold.	Medium

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that the system must satisfy in addition to its functional behavior. These requirements address performance, security, reliability, availability, scalability, and usability. Each statement is measurable and independently testable. The Java 17 / Spring Boot 3 / Spring Security / MySQL 8 / Bootstrap 5 technology stack was selected in part because it natively addresses many of the security, reliability, and scalability requirements defined below.

ID	Feature Area	Requirement Statement	Priority
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NFR-01	Performance	The system shall process and record a donation transaction within 2 seconds under normal load.	High
NFR-02	Performance	The system shall support a minimum of 500 simultaneous active user sessions without performance degradation.	Medium
NFR-03	Performance	The system shall generate any standard report within 5 seconds.	High
NFR-04	Performance	The system shall reflect aid request status updates in real time upon save.	High
NFR-05	Performance	The system shall execute all database queries with optimized indexes to prevent degradation as data volume grows.	Medium
NFR-06	Security	The system shall store all user passwords using bcrypt hashing with a minimum cost factor of 12.	High
NFR-07	Security	The system shall enforce a session timeout of 15 minutes of inactivity for all authenticated users.	High
NFR-08	Security	The system shall prevent SQL injection attacks through parameterized queries and Spring Data JPA.	High
NFR-09	Security	The system shall prevent cross-site scripting (XSS) attacks through output encoding.	High
NFR-10	Security	The system shall prevent cross-site request forgery (CSRF) attacks using Spring Security CSRF protection.	High
NFR-11	Security	The system shall restrict all direct database access to authorized server-side components only.	High
NFR-12	Security	The system shall transmit all data between client and server over HTTPS using TLS 1.2 or higher.	High
NFR-13	Security	The system shall maintain detailed access and activity logs for a minimum of 12 months.	Medium
NFR-14	Reliability	The system shall ensure zero data loss during unexpected server shutdowns through transactional database commits.	High
NFR-15	Reliability	The system shall validate data consistency before committing any transaction to the database.	High
NFR-16	Reliability	The system shall provide graceful error handling with user-friendly error messages and no unhandled system crashes.	High
NFR-17	Reliability	The system shall automatically recover unsaved form progress in the event of a session interruption.	Medium
NFR-18	Availability	The system shall maintain a minimum of 99.5% uptime on an annual basis.	High
NFR-19	Availability	The system shall perform all scheduled maintenance activities during off-peak hours.	Medium
NFR-20	Availability	The system shall notify users at least 24 hours in advance of any scheduled downtime.	Medium
NFR-21	Scalability	The system shall support at least 100,000 stored donation records without architectural redesign.	Medium

NFR-22	Scalability	The system shall support horizontal scaling through cloud-based infrastructure.	Medium
NFR-23	Scalability	The system architecture shall allow adding new modules without modifying existing core components.	Medium
NFR-24	Usability	The system interface shall follow consistent navigation and layout standards across all screens.	High
NFR-25	Usability	The system shall allow a donor to complete a donation in no more than 3 navigation steps.	High
NFR-26	Usability	The system shall comply with WCAG 2.1 Level AA accessibility standards.	High
NFR-27	Usability	The system shall display a confirmation message to the user after every critical action.	High
NFR-28	Usability	The system shall provide contextual help tooltips for key input fields and controls.	Medium
NFR-29	Usability	The system interface shall be fully responsive on desktop, tablet, and mobile devices using Bootstrap 5.	High

3. Data Information Statements

3.1 Data Collection

This section defines the data that the Charity and Aid Management System collects, stores, and manages. Each data entity is presented as a set of numbered, structured data requirement statements describing individual fields, their database-format field names, data types, whether they are required, applicable CRUD operations, and a plain-language description. These data requirements serve as the specification basis for the MySQL 8 physical database design and will be realized as tables managed through Spring Data JPA and Hibernate entity mappings. Every field supports at minimum the Read operation; writable fields support Create and Update as appropriate to their business role. Fields designated as Primary Keys (PK) are system-generated and are never entered or modified by users.

3.1.1 User Data (tbl_users)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-01	USER_ID	INT (PK)	Yes	R	Auto-generated unique identifier for each user account.
DR-02	FULL_NAME	VARCHAR(100)	Yes	C,R,U	Full legal name of the registered user.
DR-03	EMAIL_ADDRESS	VARCHAR(150)	Yes	C,R,U	Unique email; used for login and all system notifications.
DR-04	PASSWORD_HASH	VARCHAR(255)	Yes	C,U	Bcrypt-hashed password. Plain text is never stored.
DR-05	PHONE_NUMBER	VARCHAR(20)	No	C,R,U	Optional contact phone number.
DR-06	USER_ROLE	VARCHAR(30)	Yes	C,R,U	Dropdown: Donor Staff Case Manager Administrator Leadership.
DR-07	ACCOUNT_STATUS	VARCHAR(20)	Yes	C,R,U	Dropdown: Active Inactive Suspended. Default: Active.
DR-08	CREATED_DATE	DATETIME	Yes	R	System-generated timestamp when the account was created.
DR-09	LAST_LOGIN	DATETIME	No	R	System-updated timestamp of the last successful login.
DR-10	FAILED_LOGIN_COUNT	INT	Yes	R,U	Count of consecutive failed login attempts. Resets on success.

3.1.2 Campaign Data (tbl_campaigns)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-11	CAMPAIGN_ID	INT (PK)	Yes	R	Auto-generated unique identifier for the campaign.
DR-12	CAMPAIGN_TITLE	VARCHAR(150)	Yes	C,R,U	Descriptive title of the fundraising campaign.
DR-13	DESCRIPTION	TEXT(2000)	Yes	C,R,U	Detailed narrative description of campaign purpose.
DR-14	GOAL_AMOUNT	DECIMAL(10,2)	Yes	C,R,U	Target monetary amount to be raised.
DR-15	COLLECTED_AMOUNT	DECIMAL(10,2)	Yes	R,U	Running total of donations received. System-maintained.
DR-16	START_DATE	DATE	Yes	C,R,U	Date the campaign becomes active for donations.
DR-17	END_DATE	DATE	Yes	C,R,U	Date the campaign closes to new donations.
DR-18	CAMPAIGN_STATUS	VARCHAR(20)	Yes	C,R,U	Dropdown: Active Completed Closed. Auto-set when goal met.
DR-19	CATEGORY	VARCHAR(50)	No	C,R,U	Dropdown: Health Education Emergency Food General.
DR-20	CREATED_BY	INT (FK)	Yes	C,R	Foreign key to USER_ID of the staff who created the campaign.

3.1.3 Donation Data (tbl_donations)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-21	DONATION_ID	INT (PK)	Yes	R	Auto-generated unique identifier for each donation record.
DR-22	TRANSACTION_ID	VARCHAR(50)	Yes	R	System-generated unique transaction reference code.
DR-23	DONOR_ID	INT (FK)	Yes	C,R	Foreign key to USER_ID of the donating user.
DR-24	CAMPAIGN_ID	INT (FK)	Yes	C,R	Foreign key to CAMPAIGN_ID linked to this donation.

DR-25	DONATION_TYPE	VARCHAR(20)	Yes	C,R	Dropdown: Monetary In-Kind.
DR-26	DONATION_AMOUNT	DECIMAL(10,2)	No	C,R	Monetary value. Required when DONATION_TYPE = Monetary.
DR-27	ITEM_DESCRIPTION	VARCHAR(255)	No	C,R	Goods description. Required when DONATION_TYPE = In-Kind.
DR-28	PAYMENT_METHOD	VARCHAR(30)	No	C,R	Dropdown: Credit Card Debit Card Bank Transfer N/A.
DR-29	DONATION_DATE	DATETIME	Yes	R	System-generated timestamp of donation submission.
DR-30	RECEIPT_GENERATED	BOOLEAN	Yes	R,U	Flag: whether a PDF receipt has been generated. Default: false.

3.1.4 Beneficiary Data (tbl_beneficiaries)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-31	BENEFICIARY_ID	INT (PK)	Yes	R	Auto-generated unique identifier for each beneficiary.
DR-32	FULL_NAME	VARCHAR(100)	Yes	C,R,U	Full legal name of the beneficiary.
DR-33	EMAIL_ADDRESS	VARCHAR(150)	No	C,R,U	Contact email for notifications, if available.
DR-34	PHONE_NUMBER	VARCHAR(20)	Yes	C,R,U	Primary contact phone number.
DR-35	RESIDENTIAL_ADDRESS	VARCHAR(255)	Yes	C,R,U	Physical residential address of the beneficiary.
DR-36	BENEFICIARY_STATUS	VARCHAR(20)	Yes	C,R,U	Dropdown: Active Inactive. Default: Active.
DR-37	REGISTERED_BY	INT (FK)	Yes	C,R	Foreign key to USER_ID of the staff who registered the beneficiary.
DR-38	REGISTRATION_DATE	DATETIME	Yes	R	System-generated timestamp of beneficiary registration.

3.1.5 Aid Request Data (tbl_aid_requests)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-39	REQUEST_ID	INT (PK)	Yes	R	Auto-generated unique identifier for each aid request.
DR-40	BENEFICIARY_ID	INT (FK)	Yes	C,R	Foreign key to BENEFICIARY_ID requesting aid.
DR-41	AID_TYPE	VARCHAR(30)	Yes	C,R,U	Dropdown: Monetary Food Supplies Medical Other.
DR-42	DESCRIPTION	TEXT(1000)	Yes	C,R,U	Detailed description of the assistance requested.
DR-43	REQUESTED_AMOUNT	DECIMAL(10,2)	No	C,R,U	Monetary amount requested. Required for Monetary aid type.
DR-44	REQUESTED_ITEM	VARCHAR(255)	No	C,R,U	Item description. Required for in-kind aid types.
DR-45	REQUEST_STATUS	VARCHAR(20)	Yes	C,R,U	Dropdown: Pending Approved Denied Fulfilled.
DR-46	JUSTIFICATION_NOTES	TEXT(1000)	No	C,R,U	Case manager notes documenting approval or denial reason.
DR-47	DECISION_TIMESTAMP	DATETIME	No	R	System-generated timestamp of approval or denial.
DR-48	REVIEWED_BY	INT (FK)	No	R,U	Foreign key to USER_ID of the reviewing case manager.
DR-49	SUBMITTED_BY	INT (FK)	Yes	C,R	Foreign key to USER_ID of the staff who submitted the request.
DR-50	SUBMISSION_DATE	DATETIME	Yes	R	System-generated timestamp of initial request submission.
DR-51	FULFILLMENT_DATE	DATETIME	No	R,U	System-recorded timestamp when aid was delivered.
DR-52	FULFILLED_BY	INT (FK)	No	R,U	Foreign key to USER_ID of the staff who recorded fulfillment.

3.1.6 Inventory Data (tbl_inventory)

DR ID	Field Name (DB)	Data Type	Required	CRUD	Description
DR-53	ITEM_ID	INT (PK)	Yes	R	Auto-generated unique identifier for each inventory item.
DR-54	ITEM_NAME	VARCHAR(150)	Yes	C,R,U	Descriptive name of the inventory item.
DR-55	ITEM_CATEGORY	VARCHAR(50)	Yes	C,R,U	Dropdown: Food Clothing Medical Supplies Other.
DR-56	QUANTITY_ON_HAND	INT	Yes	C,R,U	Current available quantity of the item.
DR-57	LOW_THRESHOLD	INT	Yes	C,R,U	Quantity level triggering a low-stock system alert.
DR-58	UNIT_OF_MEASURE	VARCHAR(30)	Yes	C,R,U	Dropdown: Units Kg Liters Boxes Bags.
DR-59	EXPIRY_DATE	DATE	No	C,R,U	Expiry date for perishable items. Optional.
DR-60	LAST_UPDATED	DATETIME	Yes	R	System-generated timestamp of last quantity modification.

3.2 Data Security and Privacy

The system enforces data security at multiple levels to protect sensitive donor, beneficiary, and transactional information. Spring Security with JWT token authentication ensures that only verified users access system resources, and RBAC restricts what each role may view or modify. All passwords are stored using bcrypt hashing with a minimum cost factor of 12 — plain-text credentials are never persisted anywhere in the system. All client-server communication travels over HTTPS using TLS 1.2 or higher. Sensitive beneficiary records are accessible only to authorized staff, case managers, and administrators. Audit logs are maintained for all critical operations including record creation, modification, approval decisions, and login events, and those logs are retained for a minimum of 12 months.

4. Reporting Requirements

This section defines the reporting and dashboard requirements for the Charity and Aid Management System. Reports provide the organization with visibility into donation activities, campaign performance, aid distribution, and system compliance. Reporting access is restricted to authorized roles as defined by the system's role-based access control model, enforced through Spring Security @PreAuthorize annotations. All reports support filtering by date range at a minimum, and selected reports support additional dimensions such as campaign or donor. Exported formats include on-screen HTML display, CSV download for donation data, and PDF receipt generation for individual donation records.

ID	Report Name	Roles with Access	Description	Priority
RPT-01	Donation Summary Report	Administrator, Staff	Total donations by campaign, donor, and date range.	High
RPT-02	Active Campaigns Report	Admin, Staff, Leadership	All active campaigns with goal, collected amount, and % progress.	High
RPT-03	Pending Aid Requests Report	Case Manager, Administrator	All Pending requests with submission date and beneficiary details.	High
RPT-04	Fulfilled Aid Report	Admin, Staff, Leadership	All fulfilled requests with fulfillment date, staff, and resources used.	High
RPT-05	Campaign Performance Report	Administrator, Leadership	Donations per campaign over a selected period with trend data.	High
RPT-06	Aid Distribution Summary	Administrator, Leadership	Total aid distributed by type, period, and beneficiary count.	High
RPT-07	Approval Turnaround Report	Administrator	Average time from request submission to approval or denial.	Medium
RPT-08	Compliance / Audit Report	Administrator	All record modifications and key system events from the audit log.	Medium
RPT-09	Inventory Status Report	Staff, Administrator	Current inventory with items below low threshold highlighted.	Medium
RPT-10	Leadership KPI Dashboard	Leadership, Administrator	KPI dashboard: total donations, campaign progress, open vs fulfilled requests, resource availability.	High

5. System / Technical Requirements

This section defines the system and technical requirements that constrain the architecture, technology choices, and operational environment of the Charity and Aid Management System. These requirements establish the specific platform, languages, frameworks, and infrastructure components to be used in development and deployment. The selected technology stack — Java 17, Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, MySQL 8, and Bootstrap 5 — was chosen for enterprise-grade reliability, built-in security capabilities, and direct alignment with the functional and non-functional requirements defined in this document. All technical requirements are testable and must be satisfied for the system to be considered complete.

ID	Feature Area	Requirement Statement	Priority
TR-01	Platform	The system shall be deployed as a web-based application accessible through modern browsers.	High
TR-02	Platform	The system shall be hosted on a cloud platform (AWS EC2 or Render.com) supporting Java Spring Boot.	High
TR-03	Platform	The system shall implement a three-tier architecture separating presentation, application logic, and data.	High
TR-04	Frontend	The presentation layer shall be built using HTML5, CSS3, JavaScript, and Bootstrap 5.	High
TR-05	Backend	The application layer shall be built using Java 17 and Spring Boot 3.	High
TR-06	Backend	The system shall expose RESTful API endpoints using Spring MVC @RestController.	High
TR-07	Backend	The system shall use Spring Data JPA with Hibernate as the ORM layer for all database operations.	High
TR-08	Auth/Security	The system shall use Spring Security for all authentication and authorization enforcement.	High
TR-09	Auth/Security	The system shall use JSON Web Tokens (JWT) for stateless user session management.	High
TR-10	Auth/Security	The system shall enforce role-based access control using Spring Security @PreAuthorize annotations.	High
TR-11	Database	The data layer shall use MySQL 8 as the relational database management system.	High
TR-12	Database	The system shall use Hibernate entity mapping for database schema definition and management.	High
TR-13	Database	The system shall enforce referential integrity through primary key and foreign key constraints in MySQL.	High
TR-14	Hardware	Client devices shall be desktops, laptops, or tablets with a modern browser and internet access.	High
TR-15	Browser	The system shall support Chrome v110+, Microsoft Edge v110+, and Safari v15+.	High
TR-16	Network	All client-server communication shall use HTTPS with TLS 1.2 or higher.	High

TR-17	Version Control	The source code shall be managed using Git with a GitHub repository.	High
TR-18	CI/CD	GitHub Actions shall be used to automate build, test, and deployment pipelines.	Medium
TR-19	Email	The system shall integrate with an SMTP email service (e.g., SendGrid) for notification delivery.	Medium
TR-20	Logging	The system shall use SLF4J with Logback for server-side application and audit logging.	High

6. User Stories

This section presents the complete set of user stories for the Charity and Aid Management System, organized by user role. Each story follows the standard format: As a [Role], I want to [action], so that [business reason / benefit]. User stories describe system functionality from the perspective of each actor and provide the human-centered context behind the functional requirements defined in Section 2. A total of 45 user stories are defined across five roles - Donor (14), Staff (12), Case Manager (7), Administrator (8), and Organizational Leadership (4) - collectively covering all 94 functional requirements specified in this document.

ID	Role	I want to...	So that...
US-01	Donor	I want to register a new account with my name, email, and password	so that I can access the system and support campaigns.
US-02	Donor	I want to log in securely using my email and password	so that only I can access my account and donation records.
US-03	Donor	I want to browse all active fundraising campaigns	so that I can find causes that align with my interests.
US-04	Donor	I want to view the full details of a campaign including its goal and progress	so that I can make an informed donation decision.
US-05	Donor	I want to submit a monetary donation to an active campaign	so that my contribution is recorded and credited to the right cause.
US-06	Donor	I want to submit an in-kind donation to an active campaign	so that physical goods I donate are tracked in the system.
US-07	Donor	I want to download a PDF receipt after making a donation	so that I have a formal record for personal files and tax purposes.
US-08	Donor	I want to view my complete donation history	so that I can track all my contributions over time.
US-09	Donor	I want to filter my donation history by date range	so that I can quickly locate a specific donation record.
US-10	Donor	I want to filter my donation history by campaign	so that I can see all contributions I made to a specific cause.
US-11	Donor	I want to schedule a recurring donation to a campaign	so that I can contribute regularly without re-submitting each time.
US-12	Donor	I want to cancel a recurring donation I have scheduled	so that I can stop automatic contributions if my circumstances change.
US-13	Donor	I want to reset my password through a secure email link	so that I can regain access if I forget my password.
US-14	Donor	I want to update my profile contact information	so that my records remain accurate and I continue receiving notifications.
US-15	Staff	I want to create a new fundraising campaign with title, goal, dates, and category	so that the organization can begin collecting donations for a specific cause.

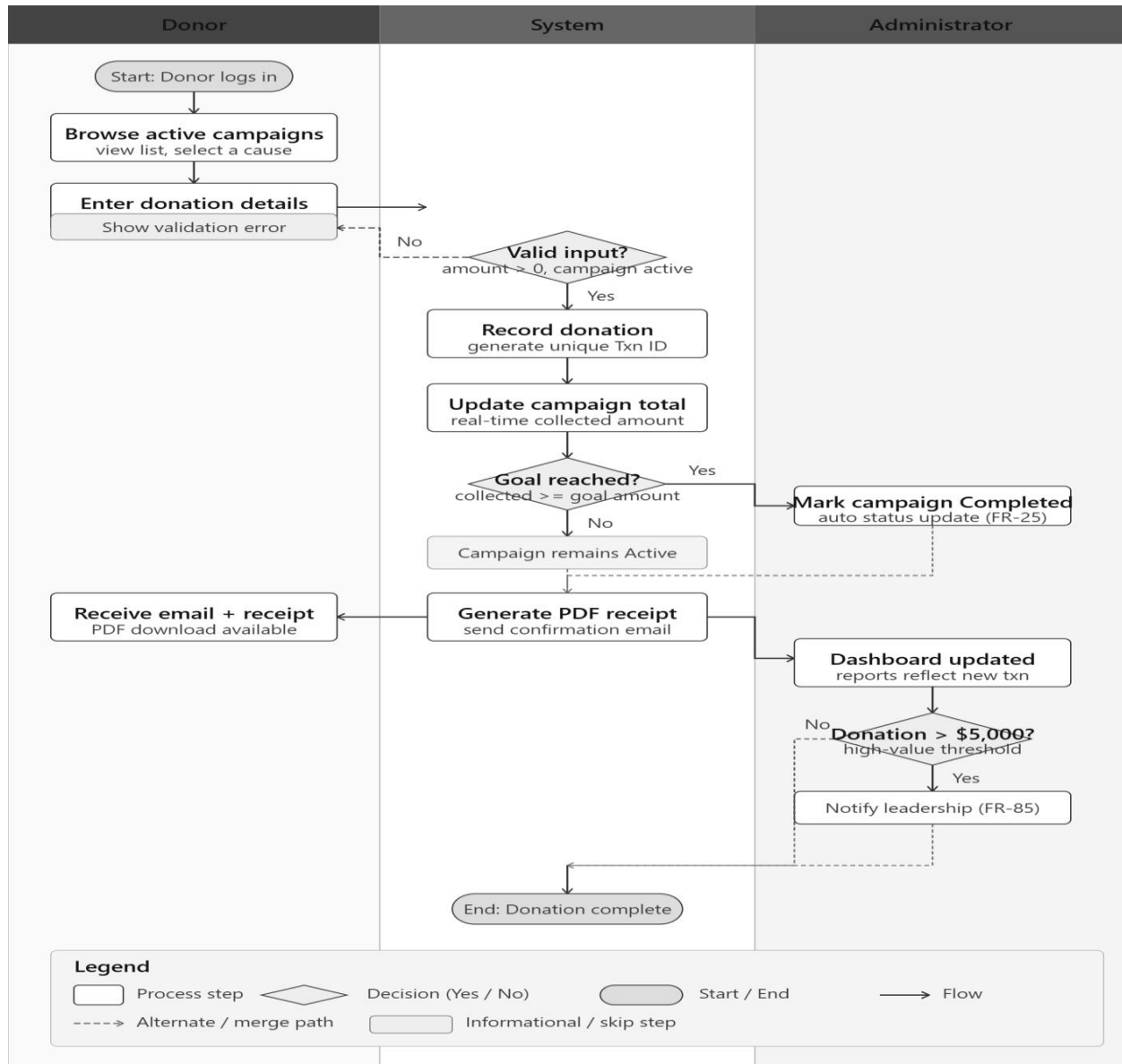
US-16	Staff	I want to update the details of an existing campaign	so that campaign information remains current and accurate for donors.
US-17	Staff	I want to close an active campaign	so that donations are no longer accepted after the campaign period ends.
US-18	Staff	I want to view all campaigns and their current status and progress	so that I have full visibility of all fundraising activities.
US-19	Staff	I want to register a new beneficiary with their name, contact, and address	so that the organization has a formal record before any aid is requested.
US-20	Staff	I want to update a beneficiary's contact information	so that their records remain accurate.
US-21	Staff	I want to view a beneficiary's profile and complete aid history	so that I understand what assistance has already been provided.
US-22	Staff	I want to record a new aid request for a registered beneficiary	so that the request enters the case manager review workflow.
US-23	Staff	I want to specify the aid type and all supporting details for a request	so that the case manager has full context to make a decision.
US-24	Staff	I want to record the fulfillment of an approved aid request	so that the system reflects that aid has been delivered.
US-25	Staff	I want to view and manage the current inventory of in-kind donation items	so that I know what physical resources are available for fulfillment.
US-26	Staff	I want to receive a low-stock alert when an inventory item falls below threshold	so that I can arrange replenishment before stock runs out.
US-27	Case Manager	I want to view a list of all pending aid requests	so that I can prioritize and work through my review queue efficiently.
US-28	Case Manager	I want to review the full details of a specific aid request	so that I have all context to make a fair and informed decision.
US-29	Case Manager	I want to approve a pending aid request	so that staff can proceed with delivering the requested aid.
US-30	Case Manager	I want to deny a pending aid request with a written justification note	so that the reason for denial is documented and transparent.
US-31	Case Manager	I want to escalate a request I cannot resolve to an administrator	so that higher-level decisions can be made when my authority is insufficient.
US-32	Case Manager	I want to view the complete status history of an aid request	so that I can see all prior actions before I review it.
US-33	Case Manager	I want to monitor the fulfillment status of all requests I have approved	so that I can ensure aid is being delivered as intended.
US-34	Administrator	I want to register new staff, case manager, or leadership user accounts	so that new team members can be onboarded into the system.
US-35	Administrator	I want to assign or change a user's role	so that access permissions are correctly set for each team member.

US-36	Administrator	I want to deactivate or suspend a user account	so that former or unauthorized users cannot access the system.
US-37	Administrator	I want to unlock an account locked after failed login attempts	so that legitimate users can regain access.
US-38	Administrator	I want to view and search the full audit log of system activities	so that I can investigate security events or compliance issues.
US-39	Administrator	I want to generate compliance and audit reports	so that the organization can meet regulatory and transparency obligations.
US-40	Administrator	I want to configure notification email templates	so that automated messages are consistent with organizational branding.
US-41	Administrator	I want to export donation data as a CSV file	so that data can be analyzed in external tools such as Excel.
US-42	Leadership	I want to view a KPI dashboard showing total donations, campaign progress, and aid statistics	so that I can monitor organizational performance at a glance.
US-43	Leadership	I want to view a campaign performance report with trend data over a selected period	so that I can identify which campaigns are most effective.
US-44	Leadership	I want to view an aid distribution summary by type and period	so that I can understand how resources are being used.
US-45	Leadership	I want to be notified when a high-value donation is received	so that I am immediately aware of significant contributions.

7. Business Process Models / Workflow Diagrams

This section describes the key business processes of the Charity and Aid Management System using structured narrative workflow descriptions. Each process identifies the participating roles, the ordered sequence of steps, and the decision points - including yes and no outcomes - that govern system behavior. These narratives correspond to the swim-lane workflow diagrams included in Section 7.1 and 7.2 . The swim-lane diagrams show each role's lane (Donor, Staff, Case Manager, Administrator, System), the sequence of actions, and all yes/no decision junctions in the full end-to-end process.

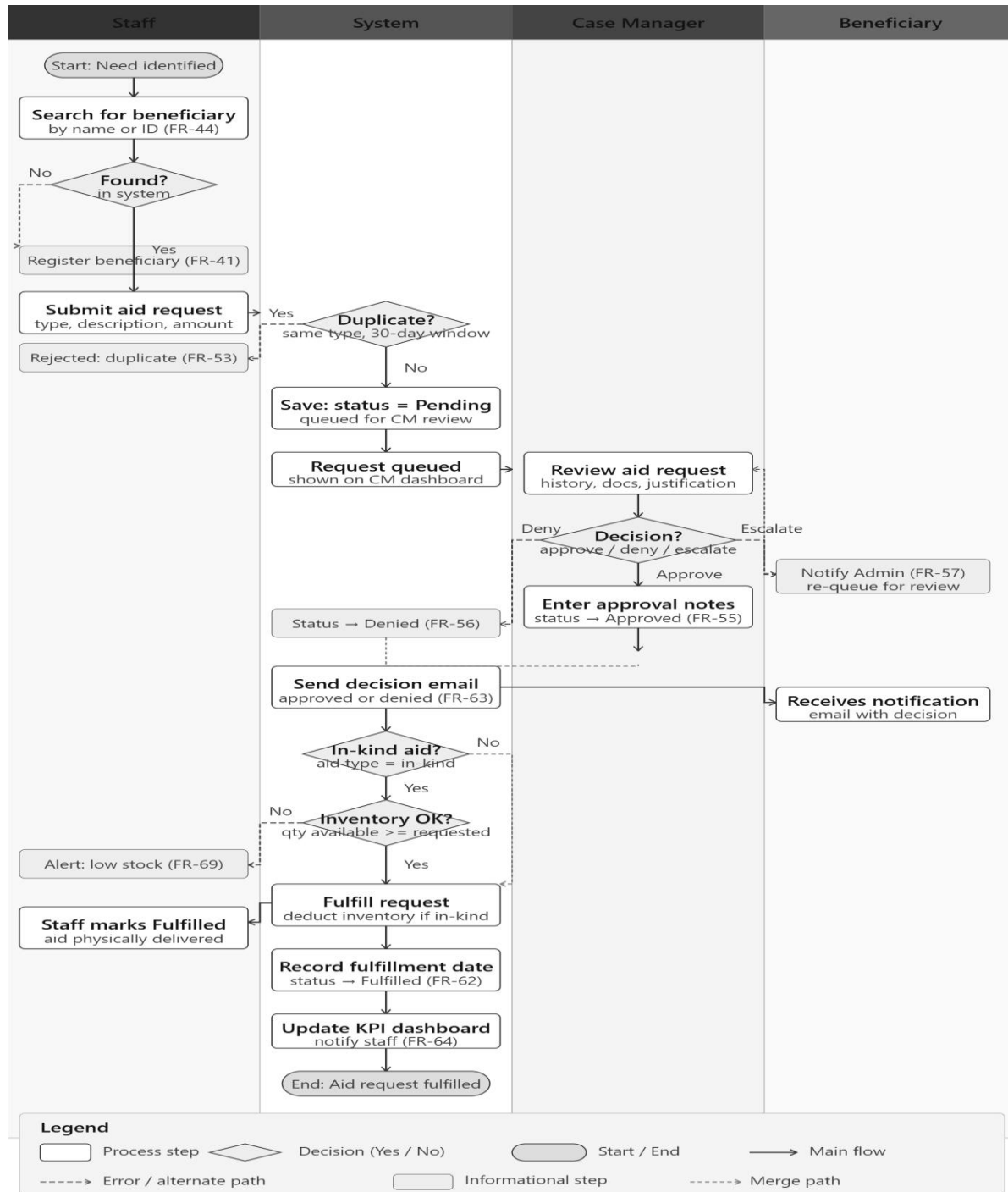
7.1 Donation Flow



Roles: Donor, Staff (monitoring), System (automated). The Donation Flow begins when a Donor logs into the system and browses the list of active campaigns. The Donor selects a campaign and reviews its description, goal amount, and current progress. Decision 1: Is the campaign Active? If No, the system blocks submission and displays a message that the campaign is closed. If Yes, the Donor proceeds to choose a donation type (Monetary or In-Kind) and enters the required details. The system validates all mandatory fields; if validation fails, the Donor is prompted to correct the

errors. Upon passing validation, the system records the donation transaction, generates a unique Transaction ID, and automatically sends a PDF receipt to the Donor by email. The campaign's Collected Amount is updated. Decision 2: Has the campaign goal now been met? If Yes, the system automatically sets the Campaign Status to Completed and sends a notification email to authorized staff. If No, the campaign remains Active. Staff members monitoring the campaign can view updated progress at any time.

7.2 Aid Request and Fulfillment Flow



Roles: Staff, Case Manager, Administrator (escalation), System (automated). The Aid Request Flow begins when a Staff member logs in and navigates to the Beneficiary module. Decision 1: Is the beneficiary already registered? If No, the Staff member registers the beneficiary before proceeding. If Yes, the Staff member proceeds to create a new aid request, specifying beneficiary, aid type, description, and requested amount or items. Decision 2: Is this a duplicate request for the same beneficiary and aid type within 30 days? If Yes, the system blocks submission and alerts the Staff member. If No, the request is saved with status Pending.

The Case Manager receives the pending request in their review queue and reviews the full request details and the beneficiary's aid history. Decision 3: Approve or Deny? If Approved, the system updates status to Approved, timestamps the decision, records the reviewing case manager, and sends an approval email to the beneficiary. If Denied, the Case Manager must enter a justification note; the system updates status to Denied, timestamps the decision, and sends a denial email. If the Case Manager cannot resolve the request, they may escalate it to an Administrator who takes the review action instead.

Once Approved, a Staff member records fulfillment. Decision 4: Is this an in-kind aid type? If Yes, the system checks inventory availability. Decision 5: Is inventory sufficient? If No, fulfillment is blocked and the Staff member is alerted to replenish stock. If Yes (or if the aid is monetary), the fulfillment is recorded, the Fulfillment Date and Fulfilled By staff member are logged, the request status is updated to Fulfilled, inventory quantities are decremented, and an automated notification is sent to the case manager confirming delivery.

8. Assumptions and Constraints

8.1 Assumptions

The following assumptions define the conditions under which the requirements of this document are valid. Any significant deviation from these assumptions may require a formal review and revision of affected requirements. Assumptions represent the agreed-upon understanding of the operating environment and stakeholder expectations at the time of writing this specification.

ID	Assumption Statement
AS-01	All users have access to a modern web browser (Chrome, Edge, or Safari) and a stable internet connection.
AS-02	Organizational staff will receive basic system training prior to go-live.
AS-03	Beneficiaries do not have direct system access; their records are managed by authorized staff on their behalf.
AS-04	The email service provider (e.g., SendGrid) will have sufficient availability and deliverability for notification use.
AS-05	The MySQL 8 database will be hosted on the same cloud platform as the Spring Boot application.
AS-06	All monetary donation amounts are recorded in US Dollars (USD).
AS-07	The system will be operated in an environment with consistent internet connectivity and cloud availability.

8.2 Constraints

The following constraints represent non-negotiable boundaries imposed on the project by schedule, approved technology decisions, organizational policy, and project scope. Constraints are not subject to waiver without formal advisor approval and revision of this document.

ID	Constraint Statement
CN-01	The system must be developed and delivered as a working prototype within the Spring 2026 semester (January–May 2026).
CN-02	The backend must be implemented using Java 17 and Spring Boot 3 as defined in the approved technology stack.
CN-03	The database must use MySQL 8 and the schema must be managed via Spring Data JPA and Hibernate.
CN-04	The system will not integrate with live payment gateways; monetary donations are recorded manually without real-time payment processing.
CN-05	Mobile native applications (iOS/Android) are out of scope; the system must be responsive on mobile browsers only.
CN-06	The system will not integrate with external charity networks or third-party CRM systems in this prototype.

CN-07	All source code must be maintained in a GitHub repository and version-controlled throughout development.
CN-08	Security implementation must use Spring Security; custom authentication frameworks are not permitted.

9. Approvals

The following individuals have reviewed this Software Requirements Specification and approve it as the authoritative baseline for the Charity and Aid Management System development project. No changes to the requirements defined herein may be made without the written consent of the approving parties.

Name	Role	Date	Signature
Shivani Pothkanuri	Author / Student		
Jeffrey Tirschman	Advisor / Instructor		
Graduate Program Director	Graduate Program Director		